

Investigation of Turbulence Created by Formula One™ Cars with the Aid of Numerical Fluid Dynamics and Optimization of Overtaking Potential

Milad Mafi

Competence Centre, transtec AG,
Tübingen, Germany

Summary

Formula One™ is the most successful racing series in the world. And with over 25 million television viewers per race it attracts many prominent sponsors who have invested over a billion euros in it, making them indispensable for Formula One™'s survival. In the last ten years the numbers of overtaking maneuvers in Formula One™ races have become rarer and rarer – changes in position now take place in the pit stop or as a result of defective parts. This has caused races to be less eventful, as reflected in falling TV viewing figures. Many sponsors have withdrawn their support and, should this trend continue, it would mean the end of Formula One™. The reason for the decreasing number of overtaking maneuvers, which were quite common in the nineteen-eighties, can be found in the aerodynamics of Formula One™ racing cars: During the age of turbo motors road holding, in racing jargon “grip” which is produced by the tires, the aerodynamics created down force resulting in extra grip. Like the tires, the proper functioning of which is restricted to a certain temperature range, the aerodynamics of a racing car depends on several factors, including the intensity of air turbulence. Laminar air flow is the basis for the efficient functioning of the wings and bodywork. The aerodynamics of modern Formula One™ cars produces a down force of roughly 25000 N, which corresponds to a gravitational force of 2.5 metric tons. This results in turbulence and eddy clouds behind the moving car. The wings of Formula One™ car fail to fulfill their function in this air layer: They produce little or no down force; instead, they create resistance which necessitates that drivers reduce their curve speeds and curve accelerations, making it impossible to execute an overtaking maneuver. FIA (Fédération Internationale de l'Automobile), has initiated a program to improve aerodynamics and minimize turbulence through changes in regulations. In this project I have investigated in which car parts create turbulences and what modifications are necessary to minimize them in order to make overtaking easier.

Keywords

automotive, CFD, FEM, FLUENT, ANSYS, TGRID, Formula One™ , F1, FSI, Compute Cluster 2003, Microsoft, transtec

1. Introduction

The 1980's are still supposed to be the best times of Formula 1. At that time, when the turbos were allowed races with a lot of overtracking manoeuvres and could be watched every second weekend, a thrilling competition was kind of usual. Drivers like Prost, Senna and Mansell and their aesthetical cars made the 1980's to an era without a comparison. One car close behind another, driving in the slipstream of the leader and the result, a lot of overtaking made the Formula One™ to the marketing machine it is today.

But in today's times the thrilling competition became to a fight of strategy. Overtaking manoeuvres take place at the pits and there is no driving close behind each other. But what has changed since the great times of Formula 1? When you look on the changes, the changes of the tires and the aerodynamics have been most important for this evolution. Cubic bolides became jets with turned-around wings.

In the following the differences between a 1980's and a modern Formula 1 car:



Source: Honda Museum, Tokyo



Source: www.mclaren.com

- Different rear wings with one element which fixes the two upper elements over the sideplates
- The front wing is placed higher and it has got more elements
- The diffuser has got more channels
- Auxiliary wings on the bodywork, the airbox and the nose were developed
- The width of the cars was reduced
- The width of the tires was reduced

All these aspects made aerodynamical grip more important than mechanical grip. Without grip the power of the 800 HP V8-engines could not be brought down on track. The aerodynamic doesn't work in turbulent flow, it needs a laminar flow for its optimal effect. Only with a huge aerodynamic grip, high corner speeds can be possible. If there is less aerodynamic grip, the lap times get slow. My work was to make a research on these aerodynamic components such as the front wing, the rear wing, the diffuser and the bargeboards and to explore what effect they have on overtaking possibilities.

1.1 Why FLUENT?

There are two possibilities to explore the aerodynamical features of a component of a Formula One™ car and especially the turbulences: the wind tunnel or computational fluid dynamics (CFD). The efficiency and the financial aspect make CFD a better solution. Even the visualisation and the accuracy are other aspects which show the advantages of CFD. New turbulence models and the increasing computing power make CFD more important, also in Formula 1. Millions were invested for high performance Clusters. The CFD Software FLUENT was chosen by 7 of 8 teams in Formula 1 (Ferrari, BMW Sauber F1 and more). The fusion with ANSYS brought even more advantages. The software package is now more effective. The kind support of the company Fluent Germany made it possible to realise this project.

Solver	FLUENT 6.3, ANSYS 11.0
Preprocessing	TGrid 4.0, GAMBIT 2.1
Postprocessing	CFD-Post

2.1 Investigated components (abstract)

To get a highly accurate result, a lot of components were first of all investigated in 2D before they were simulated in 3D. Almost every part of the external aerodynamic was investigated like this to get the most accurate results as possible.

Front wing
Rear wing
Rear wing – Diffuser - Interaction
CDG-Wing
Auxiliary winglets
Bargeboard
Diffuser
Flexible wings

2.1.1 Front wing

The main function of the front wing (Fig. 1 and Fig. 2) is to generate downforce on the front-axis. In interaction with the bargeboards and the aerodynamic at the rear of the car, it is crucial for the performance of the car. The front wing can have up to three elements and it can be up to 1400 mm wide. For this project geometries of the BMW-Sauber F1 Team were investigated.

The simulations of the 2D and 3D profiles gave a result that the front wing is only responsible for a small part of the turbulences (Fig. 3b) which have nearly no influence on the flow field. But it is very sensible for possibly changing boundary conditions: When the front wing is in a turbulent flow, it doesn't work, (Fig. 3) and loses up to 70 % of it's downforce. The most sensitive part is the first element. It loses up to 80 % downforce.

At first, the front wing was solved 2D, a tetraeder mesh with hexaeder elements in the boundary layer were used. All in all, meshes with up to 150.000 elements were used for the 2D analysis.



Fig. 1

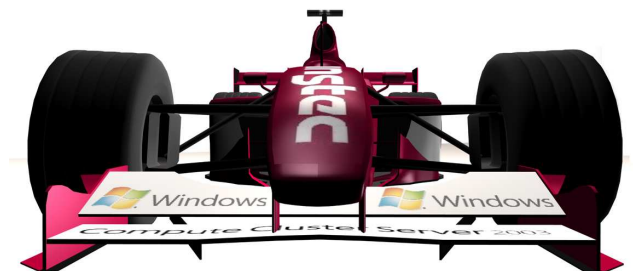


Fig. 2

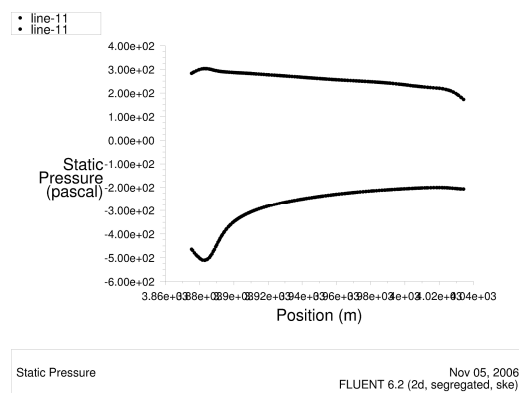


Fig. 3

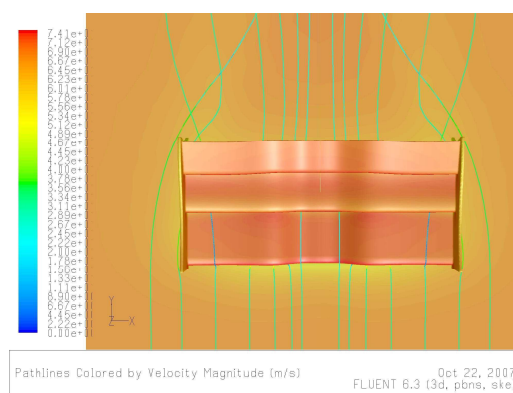


Fig. 3b

size 2D/3D	about 150 000/1 500 000
cells	tet and prism
viscous	realizable K-epsilon turbulence model

2.1.2 Rearwing

The rear wing (Fig. 4) causes about 20 % of the downforce. The regulations of the FIA say that a rear wing width of up to 1000 mm is allowed. Only up to three rear wing profiles are allowed. Especially the lower element is important for the upwash, a arduous up winding flow, which has a special interaction with the diffuser.

The rear wing was also solved by 2D first. The interaction between the rear wing and the other parts was very important for this simulation. The simulation shows that the rear wing is responsible for the main part of the turbulences (Fig. 5) especially on its edges. The upwash though is quite laminar, so the rear wing alone can not be the reason for the huge amount of turbulences behind the car.

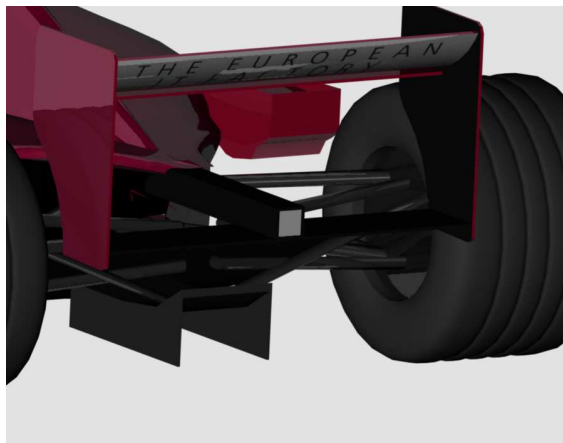


Fig. 4

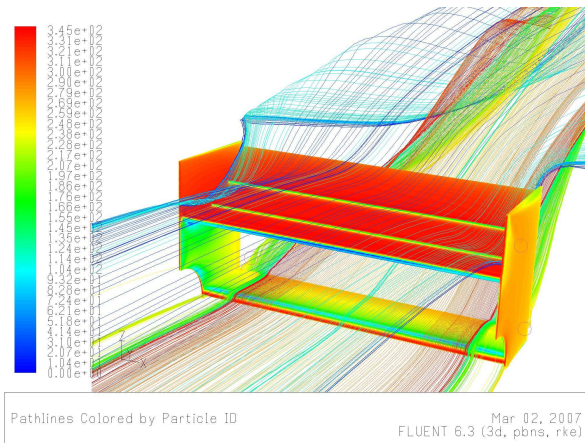


Fig. 5

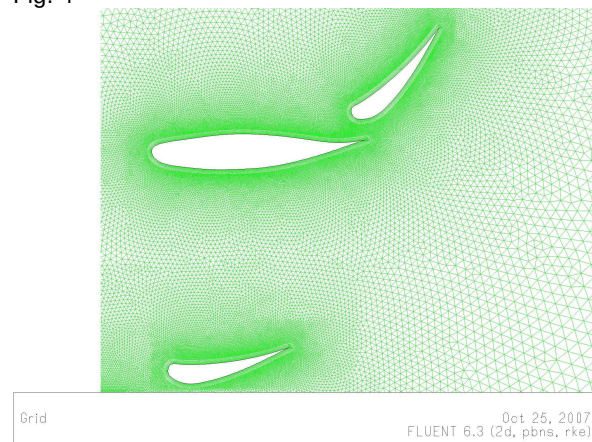


Fig. 6

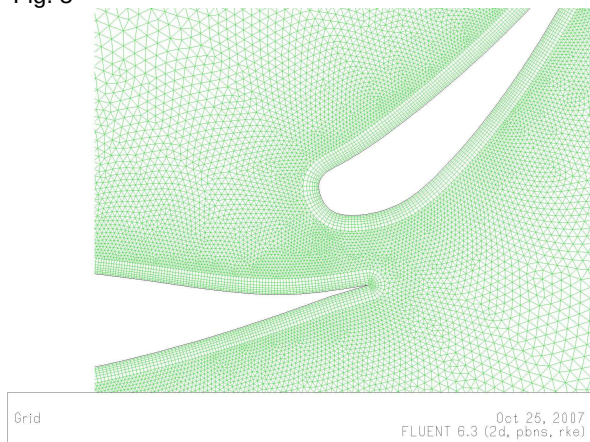


Fig. 6b

size 2D/3D	about 150 000/2 500 000
cells	tet and prism
viscous	realizable K-epsilon turbulence model

2.1.3 Rearwing-Diffuser-Interaction

The most important phenomenon is the so called „upwash“. The „upwash“ is a steeply upwards torrential current which rips the diffuser current upwards. At the same time, air diffuses from the side to the middle, which causes a further vertebration. The diffuser current itself is hardly turbulent and calms down after a few metres. In interaction with the „upwash“ however it can be carried on for several metres.

This addition is particularly revealed in the resulting complete downforce. This means that the downforce with „upwash“ which is produced by the diffuser, on the rear wing for example, is over 60 % higher than the downforce which the diffuser produces without „upwash“.

The left graphic (Fig.7) shows clearly the breaking at the diffuser and the turbulences at the backlash. The right graphic (Fig.8) shows the steadying current setting back down.

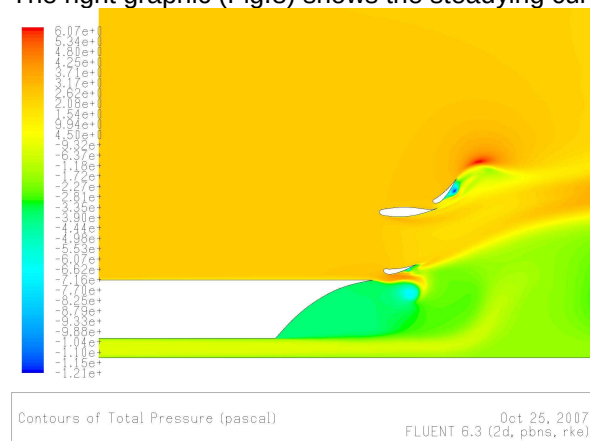


Fig. 7

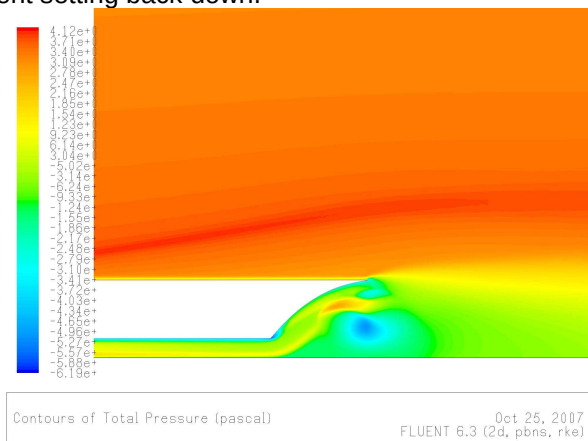


Fig. 8

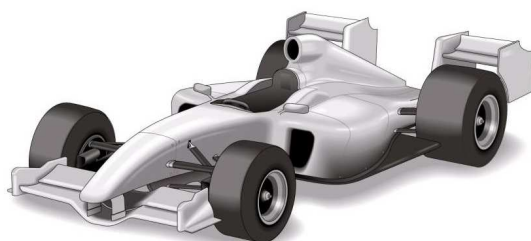
size 2D	about 210 000
cells	tet and prism
viscous	realizable K-epsilon turbulence model

2.1.4. CDG Wing

The CDG wing represents a whole aerodynamic concept. It is supposed to reduce the interaction between rear wing and diffuser. The upwash over the diffuser would dissappear.

But the concept of the CDG wing is, as shown in the simulations, not convertible because the teams would find other possibilities to create the upwash.

The simulations show that the CDG-wing would not be the best solution to interrupt this interaction. The CDG wing was completley simulated in 3D.



size 3D	about 2 500 000
cells	tet and prism
viscous	realizable K-epsilon turbulence model

Fig. 9 Car with CDG wing, Source: www.fia.com

2.1.5. Auxiliary winglets

Winglets are small additional wings with the function to control and calm down the air current on certain points. Since the beginning of the millennia, they have been entered into the designs of vehicles and mostly exist on the side box and airbox but also on the front wings of the bolides. Due to the design and amount of different winglets, it was difficult to find an abstract picture. The winglets could only be simulated in 3D. These simulations show that the winglets produce more turbulences than the front wing for example. Due to the position and amount of these aerodynamical auxiliary devices, the current is lead directly through the rear wing, which has an influence on the backlash of the vehicle.

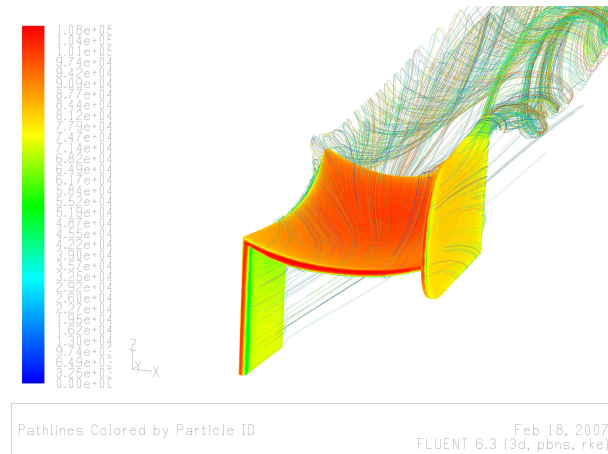


Fig. 10

size 3D	about 1 800 000
cells	tet and prism
viscous	realizable K-epsilon turbulence model

2.1.6. Bargeboard

The bargeboards were firstly installed in 1994 by Williams –Renault and primarily have the function to direct the current from the front wing. This is lead under the vehicle and around the side box. This way resistance is minimized and at the same time the current velocity at the under surface is raised. The bargeboards themselves hardly produce turbulences. Only on the upper edges of the ends and on the insides twirls form. These are however very long and disappear in the side box.

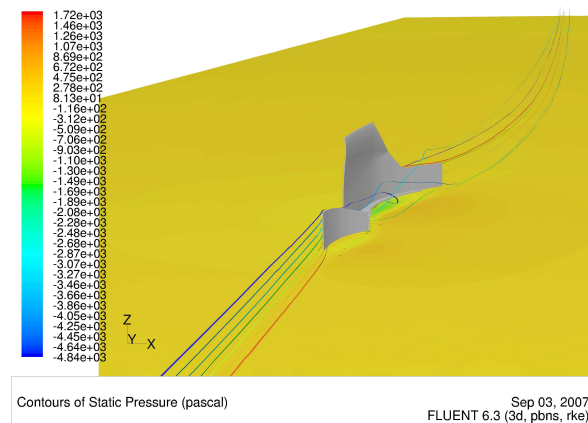


Fig. 11

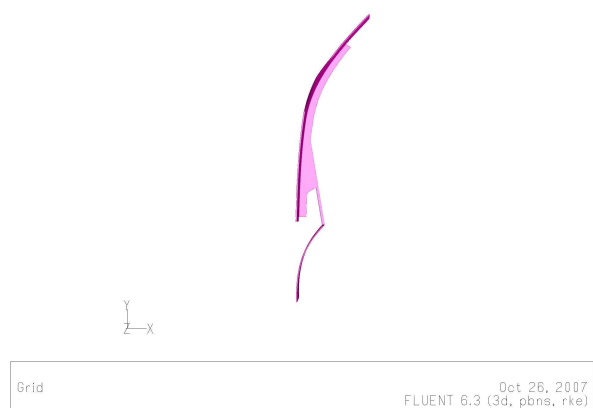


Fig. 12

2.1.7. Diffuser

In the 1970's the discovery of the "ground effect"¹ by Colin Chapman made the diffuser (Fig. 13) important to the formula 1. The diffuser is a reversed airfoil on the vehicle base – meanwhile been added by a number of channels. The diffuser produces 40 % of the complete downforce. To be able to suspend some of the effects, especially the "ground effect", prism and hexahedron elements were included to the base and the diffuser. The simulations show that the diffuser itself produces turbulences. These however have a faint influence to the backlash (Fig. 14). Particularly 2 metres behind the diffuser the turbulences are very strong and reach up to 10 metres behind the vehicle, where they however hardly take effect.



Fig. 13, Source: www.ef.org.vt.edu

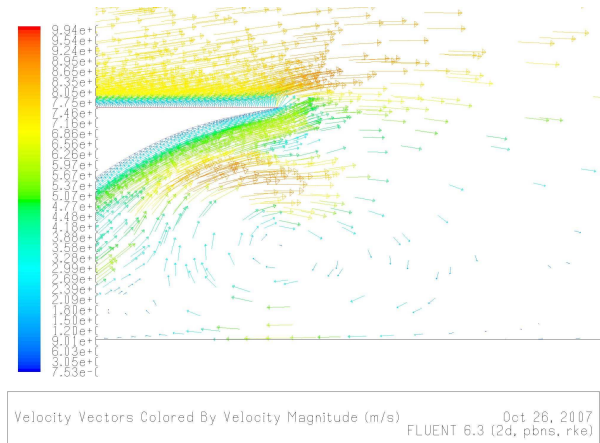


Fig.14

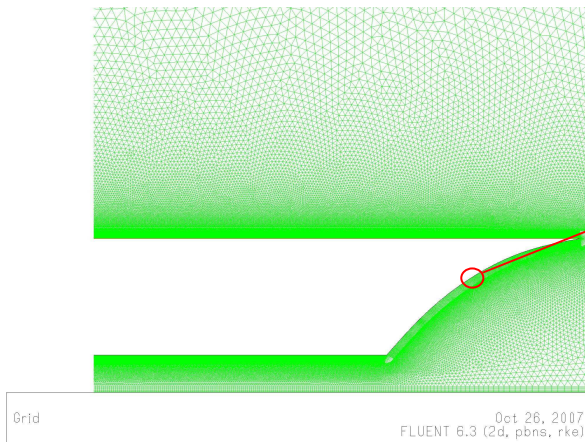


Fig. 15

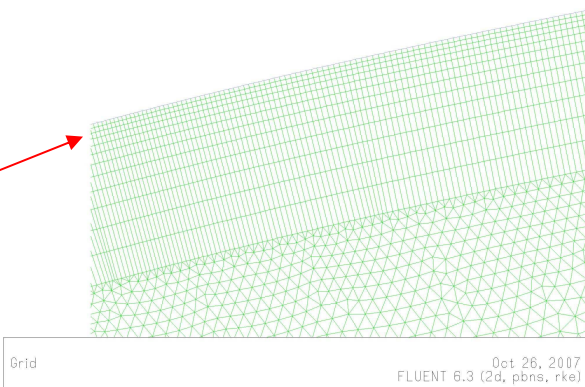


Fig. 16

size 2D	about 230 000
cells	tet and prism
viscous	realizable K-epsilon turbulence model

¹ Ground effect is a generic term describing any aerodynamic effects occurring due to a vehicle's body or appendages moving in close proximity to the ground.

2.1.8. Flexible wings

One alternative to minimize the turbulences, at least on the straight distances, are through flexible wings on the lower surfaces. Depending on the current velocity these change their profiles, distance to the ground and blade angle. Due to this, at high speed the downforce and with this the resistance and formation of turbulences can be reduced. Due to safety reasons the flexible wings were forbidden in the 60's, however those risks of material weariness through composite materials are out of date. Flexible wings and lower surfaces which rise on straight distances can achieve active contributions to minimize turbulences. With increasing blade angles the turbulences also increase behind the wing (Fig. 17; all details comparative). The same goes for the rate between downforce and turbulence intensity (Fig. 18).

Another alternative could maybe be the reintroduction of the active wheel suspension (Fig. 19).

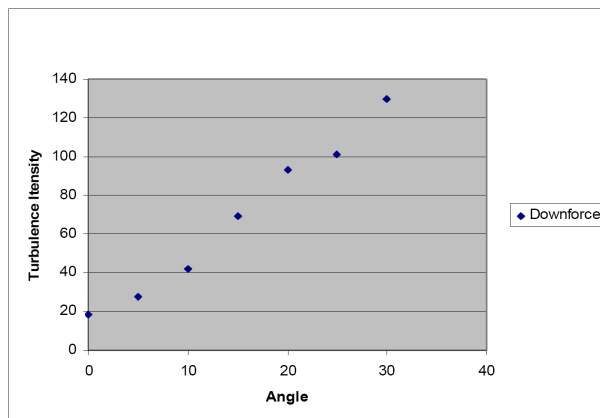


Fig. 17

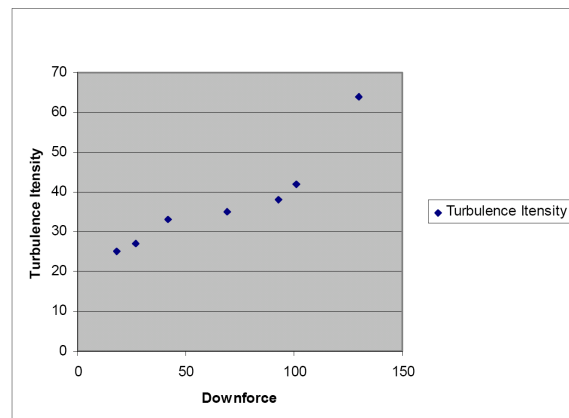
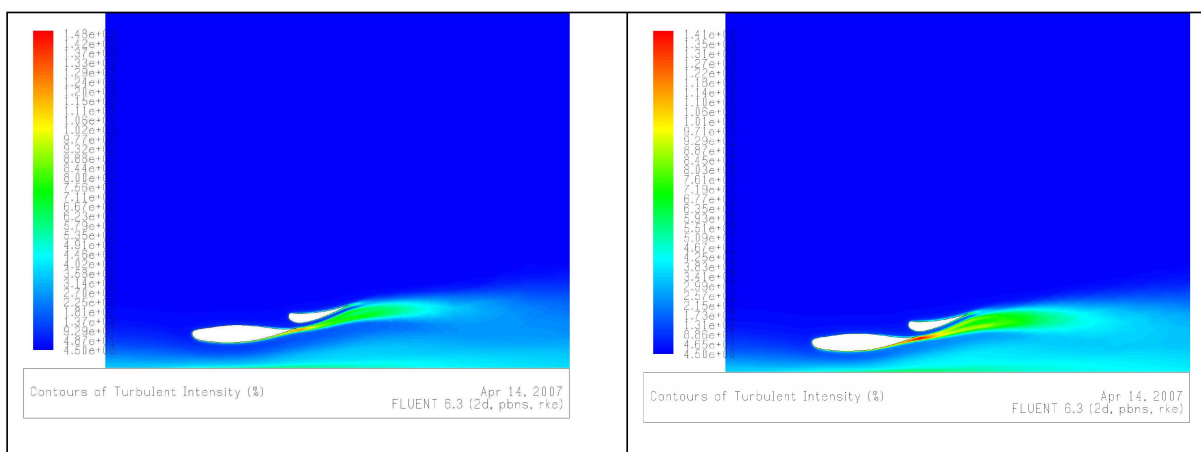


Fig. 18



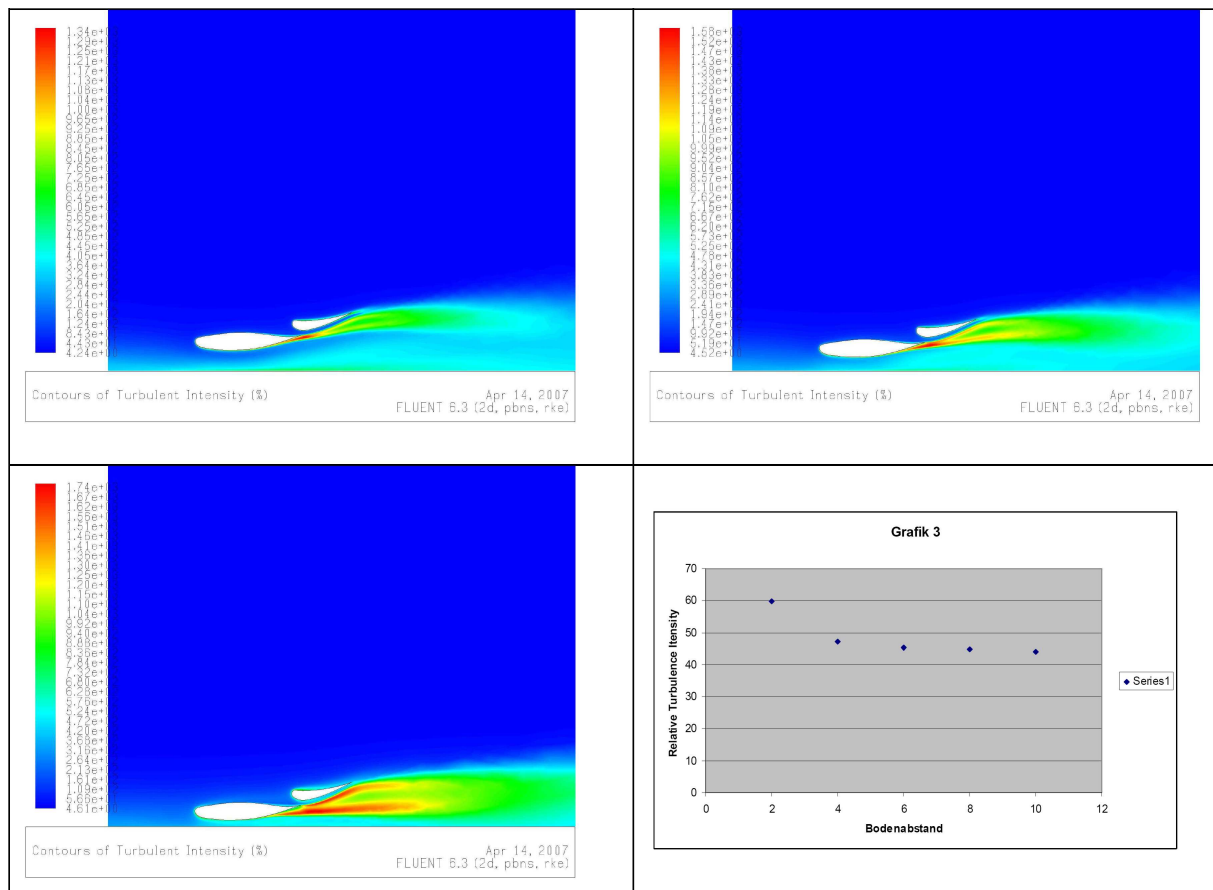


Fig. 19

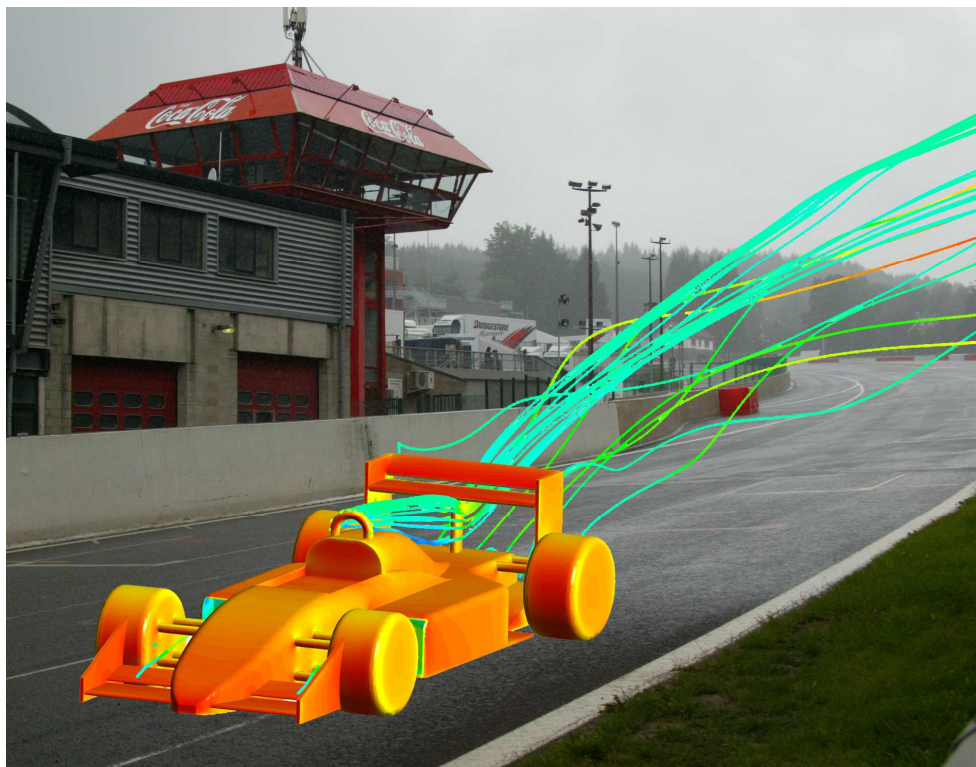


Fig. 20, Flow around a complete car (1980's)

3. Conclusions

The analyses of vehicle components with the help of FLUENT proved to be time-saving and cost-effective. Particularly the direct workflow from the CAD application to the simulation was time-saving. Due to the integrated postprocessors it was easy to analyse the results fast.

With help of the above mentioned simulation the problematical components and areas in a Formula One™ vehicle can be analysed and visualized. Especially outstanding is that the winglets and the lower rear wing element, which are in interaction with the diffuser, produce the most turbulence. These parts did not exist in the 1980's. At that time the downforce was mainly produced by the wide tires. A further alternative to minimize turbulences would be by reintroducing flexible aerodynamical components, which were forbidden after the "Concorde Agreement"². These would at least reduce the turbulences on the straight distances.

4. Outlook

Although I have been working on this project for less than a year now, I have not yet achieved my goal. I have only simulated a single 1980's car and I wish to simulate a complete car together with various scenarios and submit the results in written form at the end of 2007. However additional to this purpose, computations should also be done for smaller units such as camera enclosures, etc. and I would also like to treat flexible wings, exhaust systems and their standardisation.

Sponsors



<http://www.transtec.de>



<http://www.microsoft.com>



<http://www.ansys.com>

5. References

- [1] Mafi, M.: "Modifikationen am Heckflügel eines F1-Rennwagens zur Verbesserung der Überholmöglichkeiten", Fluent User Group Meeting, Germany 2006
- [2] Larsson, T.: "BMW Sauber F1 Team, CFD: Shaping the future of Formula 1", Fluent User Group Meeting, Germany 2006
- [3] Akanni S., Larsson T., Bienz C., "Numerical Modelling of the Aerodynamic Flow Field about a Formula One Car", Fluent User Group Meeting, Germany 2001.
- [4] Bienz C., Larsson T., Sato T., Ullbrand B. "In Front of the Grid – CFD at SAUBER PETRONAS F1 Leading the Aerodynamic Development", 1st European Automotive CFD Conference, Bingen, Germany, 2003

² The Concorde Agreement is a contract between the FIA, the Formula One teams and Formula One Administration which dictates the terms by which the teams compete in races and take their share of the television revenues and prize money.
